

Regression for Large Matrices with Joint Column and Row Reduction

Master's Thesis - Oberseminar

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Contents

Motivation

Theoretical Backbone

Joint Reduction Procedures

Simulation Study

Wrap-Up

References

Motivation

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- Increasing availability of large-scale data
 1. Rapid growth of the number of observations n
 2. High-dimensionality p of data
- Computational, statistical, and scalability challenges for traditional methods in statistical modeling

Motivation

- Increasing availability of large-scale data
 1. Rapid growth of the number of observations n
 2. High-dimensionality p of data

→ Computational, statistical, and scalability challenges for traditional methods in statistical modeling
- Limitations motivate the use of data-aware reduction techniques
 - Aim to preserve structural information of data matrix
 - ⇒ Importance-based sub-sampling methods

Theoretical Backbone

Leverage-based Importance Measures

Measures exploiting data structures to identify important columns/rows

- Random Scores

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$$s_j \sim \text{Uniform}(0, 1), \quad j = 1, \dots, d$$

→ Do not consider any data structures in quantifying importance

Leverage-based Importance Measures

Measures exploiting data structures to identify important columns/rows

- Random Scores

$$s_j \sim \text{Uniform}(0, 1), \quad j = 1, \dots, d$$

→ Do not consider any data structures in quantifying importance

- Leverage Scores

→ Based on SVD of data matrix $\mathbf{X}_{\text{Data}} = \mathbf{U}\mathbf{S}\mathbf{V}^T$

1. Row Leverage Scores

$$\ell_i^{(\text{row})} = \|(\mathbf{U}_k)_{i,\cdot}\|_2^2, \quad i = 1, \dots, n$$

2. Column Leverage Scores

$$\ell_j^{(\text{col})} = \|(\mathbf{V}_k)_{j,\cdot}\|_2^2, \quad j = 1, \dots, d$$

→ Quantify the influence of rows/columns on a matrix' low-rank structure

Further Importance Measures

Measures exploiting data structures to identify important columns/rows

- Cross Leverage Scores

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Measures exploiting data structures to identify important columns/rows

- Cross Leverage Scores

→ Based on QR-decomposition of $\tilde{\mathbf{X}} = [\mathbf{X}_{\text{Data}}, \mathbf{y}]^T = \mathbf{Q}\mathbf{R}$

$$c_{j\tilde{d}} = \langle \mathbf{Q}_{j.}, \mathbf{Q}_{\tilde{d}.} \rangle, \quad j = 1, \dots, d$$

→ Quantify the relationship between a covariate and the target variable $\mathbf{y}$

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→ Quantify the relationship between a covariate and the target variable $\mathbf{y}$

- Combined Scores

$$cs_j = p_{\text{leverage}} \cdot \frac{\ell_j^{(col)}}{\|\ell^{(col)}\|_2} + (1 - p_{\text{leverage}}) \cdot \frac{|c_{j,\tilde{d}}|}{\|\mathbf{c}_{\tilde{d}}\|_2}, \quad j = 1, \dots, d$$

→ Introduced to combine the aspects of both scores

Row Sketching

- Goal
 - Compute a $(1 + \varepsilon)$ -approximate solution $\tilde{\beta}$ of $\mathbf{y} = \mathbf{X}\beta + \varepsilon$,
 - using a substantially smaller sketching dimension/ complexity r ,
 - instead of to the original number of observations n .

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- General Approach
 - Draw sketching matrix $\mathbf{S} \in \mathbb{R}^{r \times n}$ from an appropriate distribution
 - Apply to design matrix and response vector
 - Yielding the sketched regression problem

$$(\mathbf{1}, \mathbf{S}\mathbf{X}_{\text{Data}})\beta = \mathbf{S}\mathbf{y}.$$

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- Rademacher Sketching
 - Rademacher variables taking on $\{+1, -1\}$ with equal probability
 - Draw Rademacher vectors $\mathbf{s}_i$, for $i = 1, \dots, n$ independently
 - Aggregate the rows of the data matrix according to

$$\mathbf{R} = \frac{1}{\sqrt{r}} \sum_{i=1}^n \mathbf{s}_i (\mathbf{X}_{\text{Data}})_{(i,:)} \in \mathbb{R}^{r \times d}.$$

Joint Reduction Procedures

Sequential Column-Row Reduction

Input

- 1: Data matrix $\mathbf{X}_{\text{Data}} \in \mathbb{R}^{n \times d}$, response variable $\mathbf{y} \in \mathbb{R}^n$, target rank k
- 2: Compute $c = \lceil k \log(k) \rceil$, $r = \lceil c \log(c) \rceil$

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Part 1: Importance-based Column Reduction

- 1: Compute column scores s_j for $\mathbf{X}_{\text{Data}}$
- 2: Compute probabilities $p_j = \frac{|s_j|}{\sum_{l=1}^d |s_l|}$
- 3: Apply EXPECTED(c) sampling using $\min\{1, cp_j\}$
- 4: Form reduced matrix $\mathbf{C} = \mathbf{X}_{\text{Data}} \mathbf{S} \mathbf{D}$

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Part 2: Importance-based Row Reduction

- 1: Compute row leverage scores $\ell_i^{(\text{row})}$ for $\mathbf{C}$
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- 4: Form reduced matrix $\mathbf{Z} = \mathbf{D}_r \mathbf{S}_r^\top \mathbf{C}$

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Output

- 1: Reduced matrix $\mathbf{Z} \in \mathbb{R}^{r \times c}$

Sequential Row–Column Reduction

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- 1: Data matrix $\mathbf{X}_{\text{Data}} \in \mathbb{R}^{n \times d}$, response vector $\mathbf{y} \in \mathbb{R}^n$, target rank k
- 2: Compute $r = \lceil 2k \log(d) \rceil$, $c = \lceil k \log(k) \rceil$

Sequential Row–Column Reduction

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- 1: Draw independent Rademacher vectors $\mathbf{s}_i \in \mathbb{R}^r$ for $i = 1, \dots, n$
- 2: Form sketched matrix $\mathbf{R} = \frac{1}{\sqrt{r}} \sum_{i=1}^n \mathbf{s}_i (\mathbf{X}_{\text{Data}})_{(i,:)} \in \mathbb{R}^{r \times d}$
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Part 2: Importance-based Column Reduction

- 1: Compute column scores s_j for $\mathbf{R}$
- 2: Compute probabilities $p_j = \frac{|s_j|}{\sum_{j=1}^d |s_j|}$
- 3: Apply EXPECTED(c) sampling using $\min\{1, cp_j\}$
- 4: Form reduced matrix $\mathbf{C} = \mathbf{RSD}$

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Output

- 1: Reduced matrix $\mathbf{Z} \in \mathbb{R}^{r \times c}$, reduced response $\mathbf{y}_{\text{sketched}} \in \mathbb{R}^r$

Extensions to Reduction Procedures

1. Extension to LASSO Regression

→ Introduction into row-column procedure as column reduction step

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2. Extension to Logistic Regression

→ Introduction of column-row procedure

2.1 Column reduction through importance measures

2.2 Row reduction through importance-measure specifically designed for logistic regression

Simulation Study

Dataset Generation

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$$\beta_j = \begin{cases} \rho_j, & j \in \mathcal{J}, \rho_j \sim \mathcal{N}(0, 1) \\ 0, & j \notin \mathcal{J}, \end{cases}, \mathcal{J} \text{ index set of non-zero coefficients}$$

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3. Computing the linear $\boldsymbol{\eta}$ as a simple linear combination:

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4. Computing the response vector $\mathbf{y} \in \mathbb{R}^n$

4.1 Linear regression: noisy (Gaussian) linear predictor:

$$y_i = \eta_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}\left(0, \frac{\text{Var}(\eta)}{\text{SNR}}\right)$$

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4.2 Logistic regression as a Bernoulli response:

$$y_i \sim \text{Bernoulli}(\pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

Table 1: Simulation settings for the generated datasets

Scenario	n	p	non-zero β	Signal Density	Sparsity	SNR
1	1 000	1 000	200	20%	80%	2.0
2	5 000	5 000	500	10%	90%	2.0
3	10 000	10 000	500	5%	95%	2.0

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→ Controlled progression from moderately to highly challenging high-dimensional scenarios

- Increasing the number of covariates
- Reducing the proportion of nonzero coefficients

→ Varying degrees of sparsity and complexity

Study Setup

Analysis of different desired target ranks k of reduced data matrix

$$k \in \{10, 20, 25, 50, 100, 200, 300, 400, 500\}$$

→ Perform analysis approach for each k

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Analysis Approach

- Loop over 10 seeds
 - Loop over 10 different datasets

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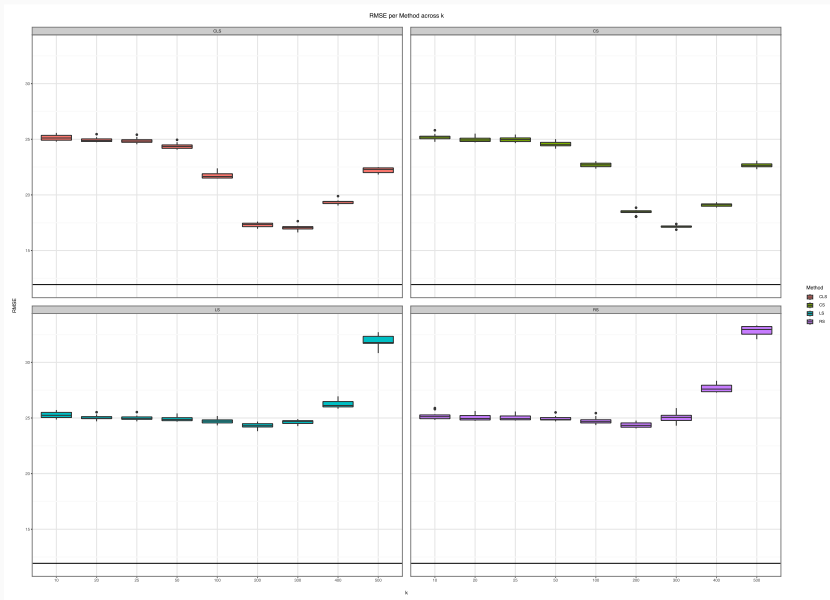
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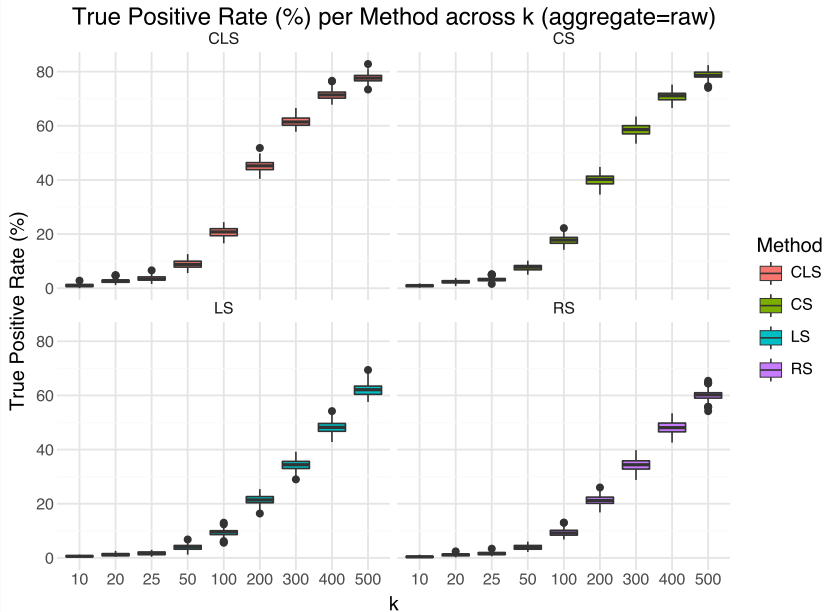
Analysis Approach

- Loop over 10 seeds
 - Loop over 10 different datasets
 1. Creation of an 80/20 train-test split
 2. Performing joint data reduction procedure
 3. Apply linear/logistic regression model
 4. Calculate benchmark model based on synthetic coefficient vector
 - Aggregate results by median

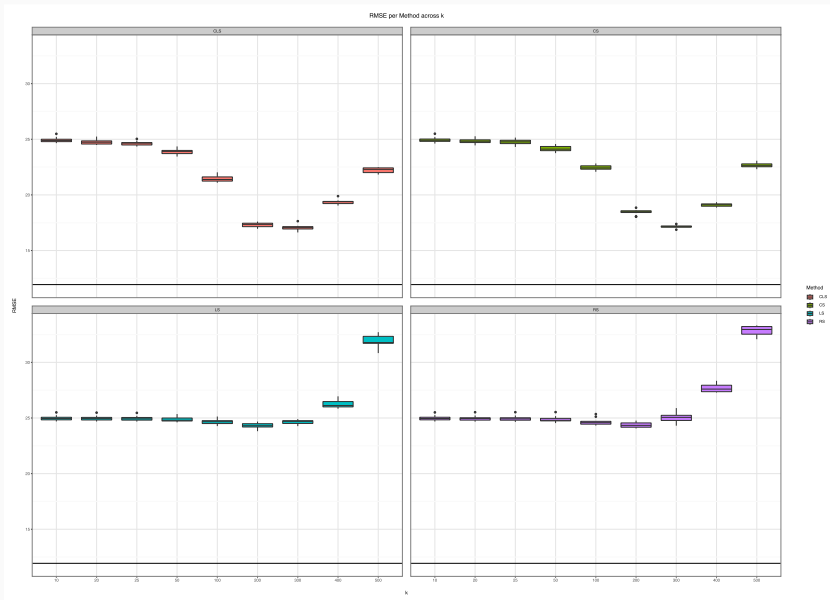
Results - Sequential Column-Row Reduction



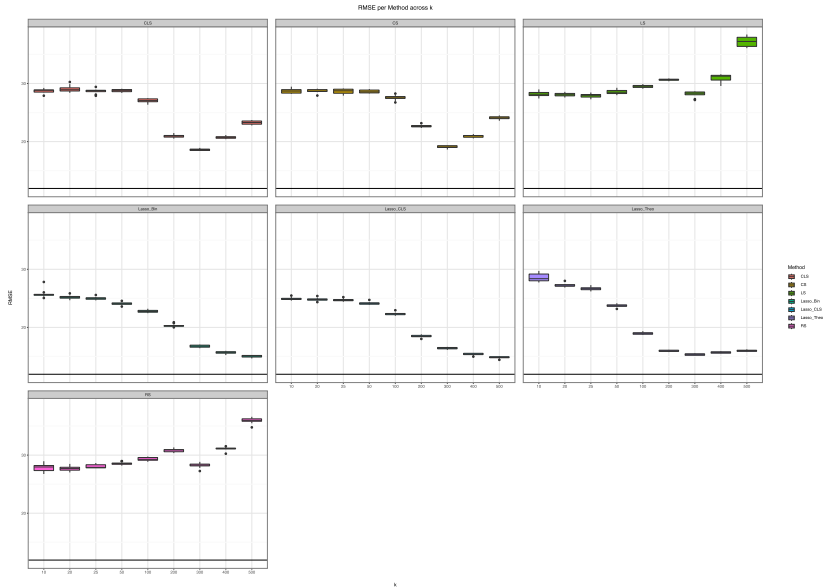
Results - Screening Performance



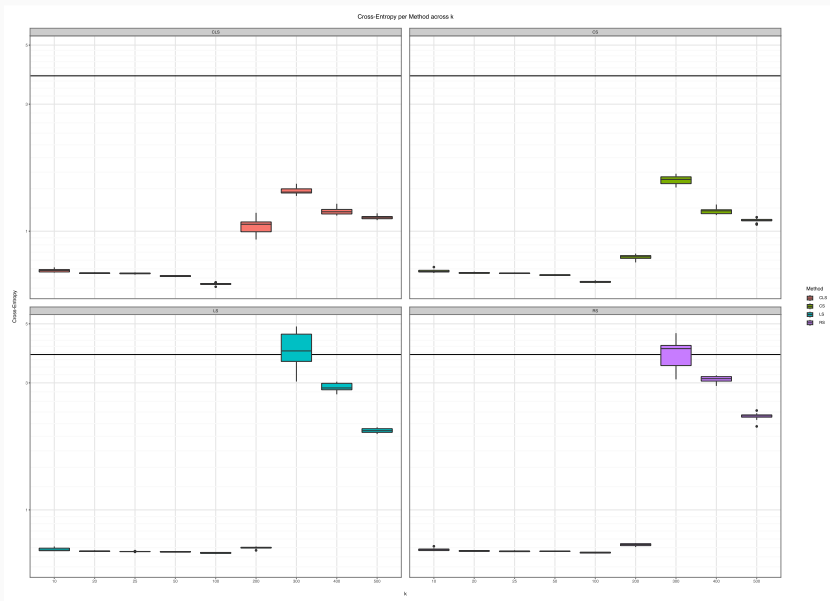
Results - Only Column Reduction



Results - Sequential Row-Column Reduction



Results - Extension to Logistic Regression



Wrap-Up

- Comparison between Importance Measures/Scores
 - For all procedures: Cross-Leverage Scores are superior
 - Computation time of Cross-Leverage Scores reasonably low, too

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- Comparison of Column-Row Reduction Order
 - Row-first Pipeline is superior and achieves lower RMSE
- Extension to Logistic Regression
 - Reduction techniques do not perform as well as in OLS scenario
 - Cross-Leverage Scores still most successful

Thank you for your attention!

Questions?

References

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Backup

EXPECTED(c) Algorithm

Part 1: Column Reduction

- 1: Compute column scores s_j for $\mathbf{X}_{\text{Data}}$ and probabilities: $p_j = \frac{|s_j|}{\sum_{l=1}^d |s_l|}$ for $j = 1, \dots, d$
- 2: Initialize sampling matrix $\mathbf{S} = \mathbf{0} \in \mathbb{R}^{d \times c}$, rescaling matrix $\mathbf{D} = \mathbf{0} \in \mathbb{R}^{c \times c}$
- 3: $t \leftarrow 1$
- 4: **for** $j = 1, \dots, d$ **do**
- 5: Sample column j with probability $\min\{1, cp_j\}$
- 6: **if** column j is selected **then**
- 7: $S_{jt} \leftarrow 1$
- 8: $D_{tt} \leftarrow \frac{1}{\sqrt{cp_j}}$
- 9: $t \leftarrow t + 1$
- 10: **end if**
- 11: **end for**
- 12: $\mathbf{C} \leftarrow \mathbf{X}_{\text{Data}} \mathbf{S} \mathbf{D}$

Column–Row Reduction for Logistic Regression

Input

- 1: Data matrix $\mathbf{X}_{\text{Data}} \in \mathbb{R}^{n \times d}$, target rank k
- 2: Compute $c = \lceil k \log(k) \rceil$

Part 1: Column Reduction

- 1: Compute column scores s_j for $\mathbf{X}_{\text{Data}}$
- 2: Compute probabilities $p_j = \frac{|s_j|}{\sum_{l=1}^d |s_l|}$
- 3: Apply EXPECTED(c) sampling using $\min\{1, cp_j\}$
- 4: Form reduced matrix $\mathbf{C} = \mathbf{X}_{\text{Data}} \mathbf{S} \mathbf{D}$

Part 2: Row Reduction

- 1: Compute data-dependent complexity μ and $r = \lceil c\mu \log(c\mu) \rceil$
- 2: Compute ℓ_1 and ℓ_2 row leverage scores $\ell_{1,i}$ and $\ell_i^{(\text{row})}$ for $\mathbf{C}$
- 3: Compute total scores as $\mu\ell_{1,i} + \ell_i^{(\text{row})} + \mu d \frac{1}{n}$
- 4: Compute probabilities $q_i = \frac{\mu\ell_{1,i} + \ell_i^{(\text{row})} + \mu d \frac{1}{n}}{\sum_{l=1}^n \mu\ell_{1,l} + \ell_l^{(\text{row})} + \mu d \frac{1}{n}}$
- 5: Sample r rows without replacement according to q_i
- 6: Form reduced matrix $\mathbf{Z} = \mathbf{D}_r \mathbf{S}_r^\top \mathbf{C}$

Output

- 1: Reduced matrix $\mathbf{Z} \in \mathbb{R}^{r \times c}$

Complexity Measure μ

- Sublinear sketches or coresets for logistic regression do not exist
- Introduction of a complexity measure $\mu(\mathbf{X})$
 - Describes how well the data matrix of a logistic regression problem can be compressed
 - Defined as

$$\mu_p(\mathbf{X}) = \sup_{\beta \in \mathbb{R}^d \setminus \{0\}} \frac{\sum_{x_i \beta > 0} |x_i \beta|^p}{\sum_{x_i \beta < 0} |x_i \beta|^p},$$

- Case of logistic regression, $p = 1$
- $\mu_1(\mathbf{X})$ is generally unbounded, but
 - $\mathbf{X}$ being described as μ -complex if $\mu_1(\mathbf{X}) \leq \mu \leq \infty$
 - many real-world data sets exhibit small values of $\mu_1(\mathbf{X})$, making effective compression possible

ℓ_1 Row Leverage Scores for Logistic Regression

- ℓ_1 row leverage scores

$$\ell_i^{(1)} = \|\mathbf{Q}_{i,\cdot}\|_1 \text{ for } i = 1, \dots, n,$$

- Based on QR-decomposition of the input matrix $\mathbf{C} = \mathbf{Q}\mathbf{R}$ coming from previous column reduction

Cross-Leverage Score Details

- Formal definition based on the augmented data matrix

$$\tilde{\mathbf{X}} = [\mathbf{X}_{\text{Data}}, \mathbf{y}]^T \in \mathbb{R}^{\tilde{d} \times n}, \quad \tilde{d} = d + 1,$$

- Defined as the off-diagonal elements of the hat matrix of $\tilde{\mathbf{X}}$
- Quantifying the geometric relationship between each covariate and the target variable
- Using the QR-decomposition to get better computation

$$\tilde{\mathbf{X}} = \mathbf{Q}\mathbf{R}$$

- $\mathbf{Q} \in \mathbb{R}^{\tilde{d} \times \tilde{d}}$ is an orthonormal basis of the column space of $\tilde{\mathbf{X}}$
- $\mathbf{R} \in \mathbb{R}^{\tilde{d} \times n}$ an upper-triangular matrix, representing $\tilde{\mathbf{X}}$ in the basis defined by $\mathbf{Q}$

Cross-Leverage Score Details

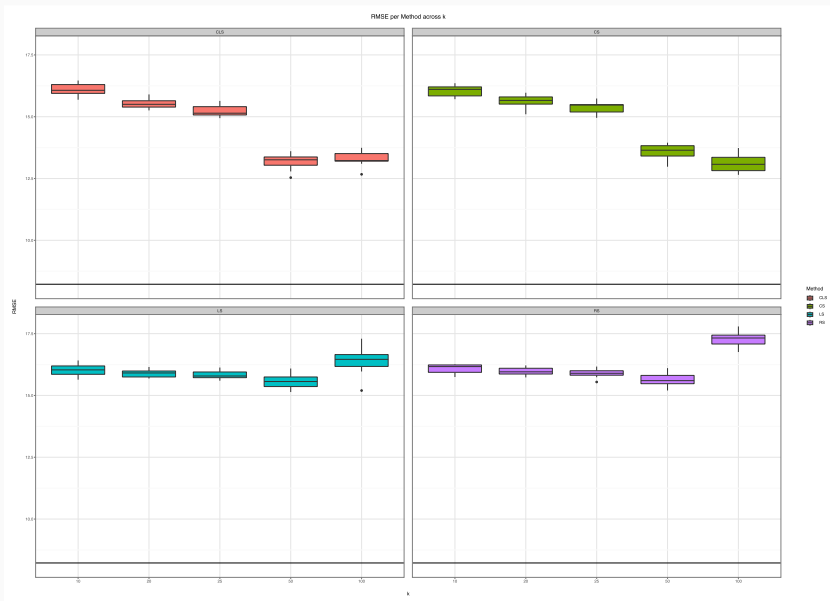
- Imputation into hat matrix

$$\begin{aligned} H &= \tilde{X}(\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \\ &= QR(R^T Q^T QR)^{-1} R^T Q^T \\ &= QRR^{-1}(R^T)^{-1} R^T Q^T \\ &= QQ^T, \end{aligned}$$

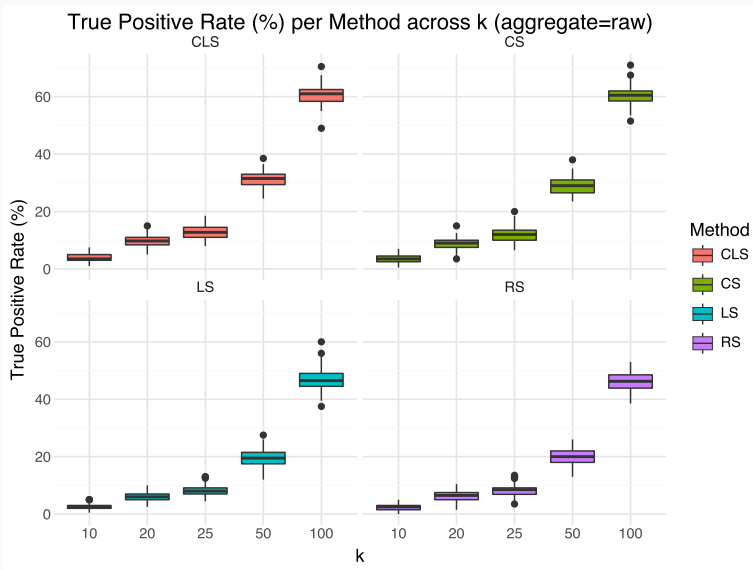
- computing H is computationally expensive due to the matrix multiplication of Q and Q^T
- more efficiently using only the corresponding rows of Q

$$c_{j\tilde{d}} = \langle Q_{j.}, Q_{\tilde{d}.} \rangle$$

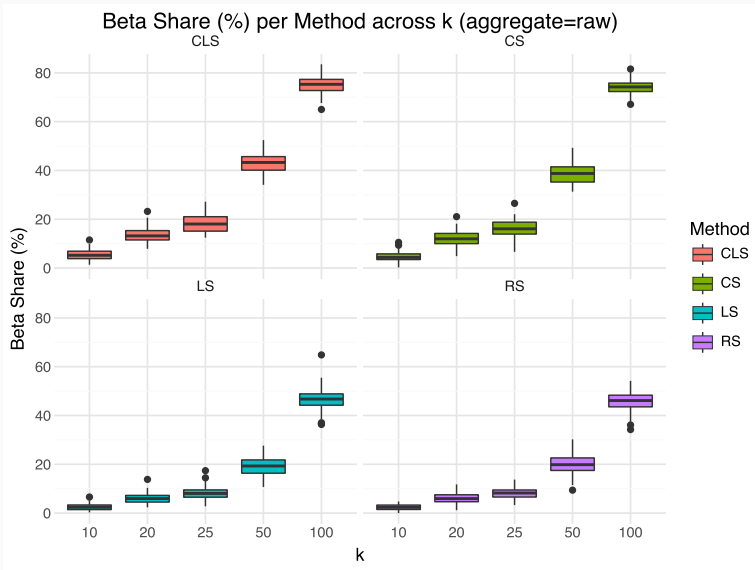
Results - Sequential Column-Row Reduction (1 000)



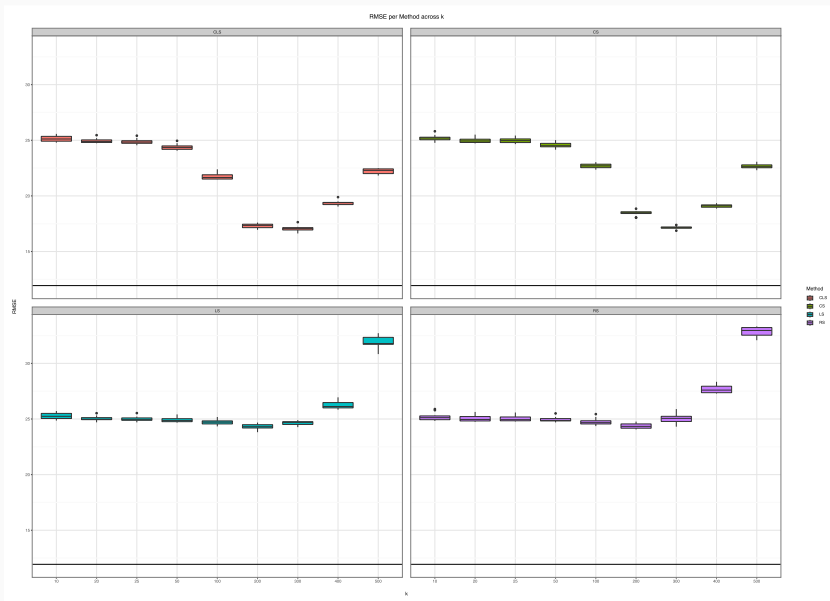
Results - Screening Performance (1 000)



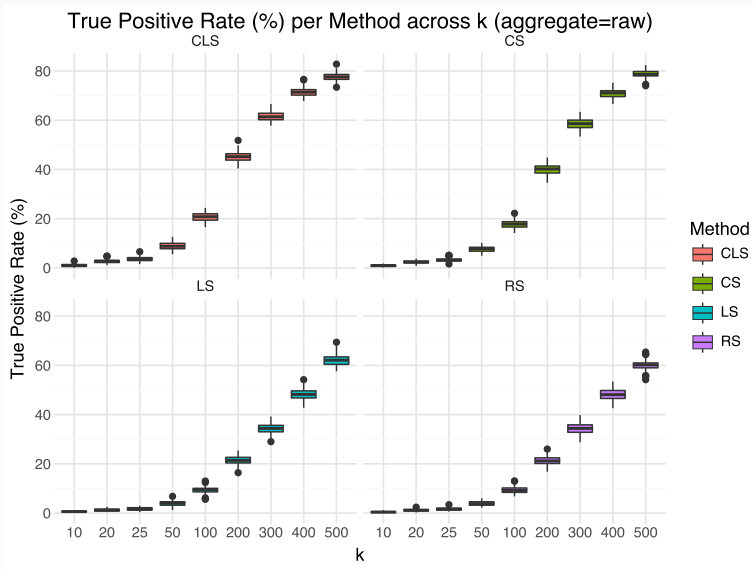
Results - Screening Performance (1 000)



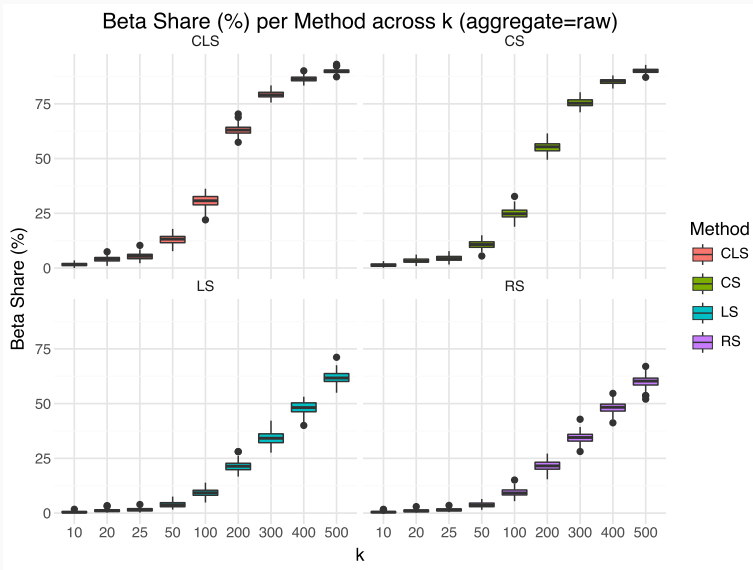
Results - Sequential Column-Row Reduction (5 000)



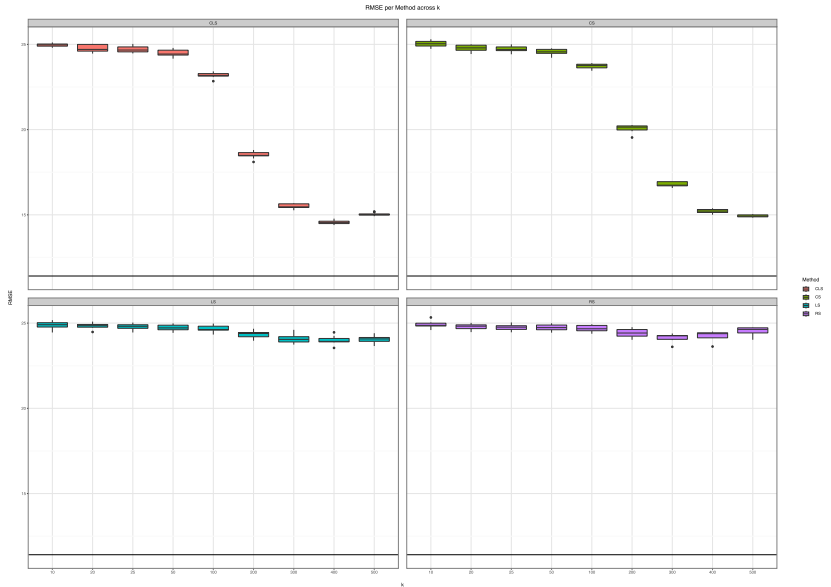
Results - Screening Performance (5 000)



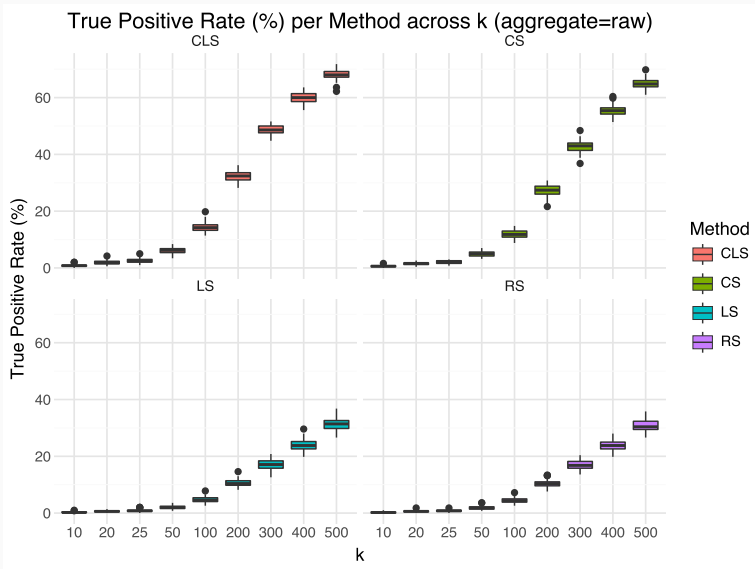
Results - Screening Performance (5 000)



Results - Sequential Column-Row Reduction (10 000)



Results - Screening Performance (10 000)



Results - Screening Performance (10 000)

